

Workshop Sheet - Week 3

Groups, homomorphisms and the first isomorphism theorem. We use the proof tactics from last week. Here is some useful notation.

<code>[Group G]</code>	Equip the type G with a group structure.
<code>f : G →* H</code>	A group homomorphism $f : G \rightarrow H$.
<code>K : Subgroup G</code>	A subset of G equipped with proofs of subgroup closure.
<code>f.ker, f.range</code>	The kernel and image, both subgroups.
<code>e : G ≃* H</code>	A group isomorphism $G \cong H$.

Useful lemmas. For group calculations: `mul_assoc`, `mul_one`, `one_mul`, `inv_mul_cancel`, `mul_inv_cancel`, `mul_inv_rev` and `pow_add`. For homomorphisms, `map_one f`, `map_mul f a b` and `map_inv f a` express that f preserves the identity, multiplication and inverses. Definitions and quotient facts are supplied in the Lean file.

1. *Group calculations.* Let G be a group and $a, b \in G$. Prove:

(a) $a \cdot 1 = a$.

(b) $a^{-1}a = 1$.

For the remaining parts instead of `simp` use `rw` with the lemmas given in the preamble.

(c) $a^{-1}(ab) = b$.

(d) $(ab)^{-1} = b^{-1}a^{-1}$.

(e) $(ab)b^{-1} = a$.

(f) $a^2a^3 = a^5$.

2. *Subgroups.* Let $S \subseteq G$ be nonempty, and suppose $ab^{-1} \in S$ whenever $a, b \in S$.

(a) Prove that $1 \in S$.

(b) If $a \in S$, prove that $a^{-1} \in S$. (Hint: use (a).)

(c) If $a, b \in S$, prove that $ab \in S$. (Hint: use (b).)

(d) Prove and formalize the following theorem, using (a)–(c).

Theorem (Subgroup test). *Let G be a group and $S \subseteq G$ a nonempty subset. If $ab^{-1} \in S$ for all $a, b \in S$, then S is a subgroup of G .*

3. *Kernels and images.* Let $f : G \rightarrow H$ be a group homomorphism.

(a) Prove that $f(ab^{-1}) = f(a)f(b)^{-1}$.

(b) Use Question 2 to show that $\{a \in G \mid f(a) = 1\}$ is a subgroup of G .

(c) Use Question 2 to show that $f(G)$ is a subgroup of H .

(d) Prove that $a^{-1}b \in \ker f$ if and only if $f(a) = f(b)$.

4. *The first isomorphism theorem.* Write $[a]$ for the coset of a in $G/\ker f$. The quotient group and the induced homomorphism $\bar{f} : G/\ker f \rightarrow \text{im } f$, $[a] \mapsto f(a)$, are supplied.

(a) Prove that $[a] = [b]$ if and only if $f(a) = f(b)$, using Question 3(d).

(b) Prove that $\bar{f}$ is injective. (Hint: choose representatives and use (a).)

(c) Prove that $\bar{f}$ is surjective onto $\text{im } f$.

(d) Prove and formalize the following theorem, using (b) and (c).

Theorem (First isomorphism theorem). *Let $f : G \rightarrow H$ be a group homomorphism. Then the map $[a] \mapsto f(a)$ induces an isomorphism $G/\ker f \cong \text{im } f$.*

5. *An application: the Chinese remainder theorem.* Recall $a \sim_n b$ if $n \mid (b - a)$, and write $[a]_n$ for the class of a in $\mathbb{Z}/n\mathbb{Z}$. Consider the additive homomorphism

$$F : \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}, \quad a \longmapsto ([a]_2, [a]_3).$$

(a) Show that $[a]_n = [b]_n$ if and only if $a \sim_n b$, using last week's relation.

(b) Prove that $\ker F = 6\mathbb{Z}$.

(c) Prove that F is surjective. (Hint: $3a + 4b$ maps to $([a]_2, [b]_3)$.)

(d) Apply the first isomorphism theorem in additive notation to deduce $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.

(e) Challenge: Prove and formalize the following theorem, by generalising (b)–(d).

Theorem (Chinese remainder theorem). *Let $m, n > 0$ be coprime integers. Then $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/(mn)\mathbb{Z}$ as additive groups. (Hint: use Bézout's identity to prove surjectivity.)*

In Lean, $\mathbb{Z}/n\mathbb{Z}$ is `ZMod n`; additive homomorphisms and isomorphisms use `→+` and `≃+`.